National Landslide Risk Reduction Strategy & SOP (Part 1)

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Section 1.1: Geological Hazard Overview in Indian Mountain Regions

Approximately 12.6 percent of the land area of India is prone to landslide hazards, encompassing the Himalayas, the Western Ghats, the Nilgiris, and parts of the Eastern Ghats. Landslides are triggered primarily by intense monsoon downpours, tectonic seismic shaking, and uncontrolled slope cutting for highway expansion.

The Geological Survey of India (GSI) is the nodal agency for landslide susceptibility mapping, rainfall threshold monitoring, and regional landslide early warnings.

Section 1.2: Pre-Failure Slope Indicators and Early Warning Signs

Catastrophic slope failures rarely occur without subtle physical warning signs on the hillside:

- Appearance of new widening tension cracks in the soil, retaining walls, paved driveways, or building foundation joints.
- Doors and window frames sticking or jamming for the first time due to subtle structural shearing of the building frame.
- Sudden emergence of new water springs, saturated ground seeps, or unusually muddy water gushing from slope drainage pipes.
- Leaning of utility power poles, fence posts, or hillside pine and oak trees (known as 'drunken trees').
- Faint rumbling sounds, rolling gravel noises, or unusual snapping of tree roots heard from uphill slopes.

National Landslide Risk Reduction Strategy & SOP (Part 2)

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Section 2.1: Active Debris Flow Escape Protocol: Lateral Sprint

When an active debris flow, mudslide, or rock avalanche is in motion, escape response time is measured in seconds:

- RUN LATERALLY: Sprint sideways across the slope, perpendicular to the flow line of the debris channel, aiming for stable bedrock ridges or elevated ground shoulders.
- NEVER RUN DOWNHILL in the path of the slide. Debris avalanches funnel down natural gullies and river channels at velocities exceeding 40 to 80 km/h, far outstripping human running speed.
- Drop heavy baggage, tools, and unnecessary gear immediately to preserve mobility and running agility.
- If you are trapped outdoors and escape is physically impossible, take refuge behind a sturdy barrier and assume the compact fetal curl position: tuck your knees to your chest and tightly interlock your hands behind your neck to protect your skull and cervical spine.

Section 2.2: Mountain Road & Highway Corridor Precautions

Traveling along mountain ghat roads and border passes requires heightened vigilance during monsoon season:

- Trickling pebbles and small rock fragments falling from a road cutting indicate that slope shear failure has begun directly above. Stop immediately and retreat to a wide, stable section of the road.
- Never attempt to drive across freshly fallen mud, gravel, or timber debris; the additional vehicle weight can trigger secondary slide displacement.
- Stay clear of culverts and drainage bridges where debris dams can form and breach without warning.
- In the event of a road blockage, park close to the inner mountain cut side, away from the valley drop-off, and leave hazard flashers engaged.

National Landslide Risk Reduction Strategy & SOP (Part 3)

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Section 3.1: Landslide Dams & Downstream Flash Flood Threats

A critical secondary hazard of large-scale landslides is the temporary damming of mountain rivers and streams:

- When debris blocks a river valley, an artificial impoundment lake forms upstream. Saturated loose mud dams have zero structural cohesion and can breach catastrophically within hours.
- If river flow downstream drops abruptly to a trickle during heavy rain, suspect an upstream landslide dam immediately.
- EVACUATE DOWNSTREAM RIVER VALLEYS IMMEDIATELY: Move at least 30 to 50 meters vertically above the valley floor onto stable mountain flanks.
- Report the river blockage immediately to 112, District Emergency Operation Centres (DEOC), and the Central Water Commission.

Section 3.2: Post-Landslide Safety and Secondary Slide Mitigation

- Stay completely clear of the slide scarp and runout zone. Vibration, rainfall, and unstable overhangs frequently trigger secondary slides on freshly sheared surfaces.
- Check for severed domestic gas lines, ruptured water pipes, and fallen electrical transmission cables.
- Do not drink untreated water from streams near the landslide area due to heavy silt contamination and upstream carcasses.
- Allow trained search and rescue teams (NDRF, SDRF, Army) equipped with sniffer dogs and acoustic listening devices to handle search operations in unstable mud deposits.